

# Research Methodology in Computer Science

# WRITING A RESEARCH PROPOSAL

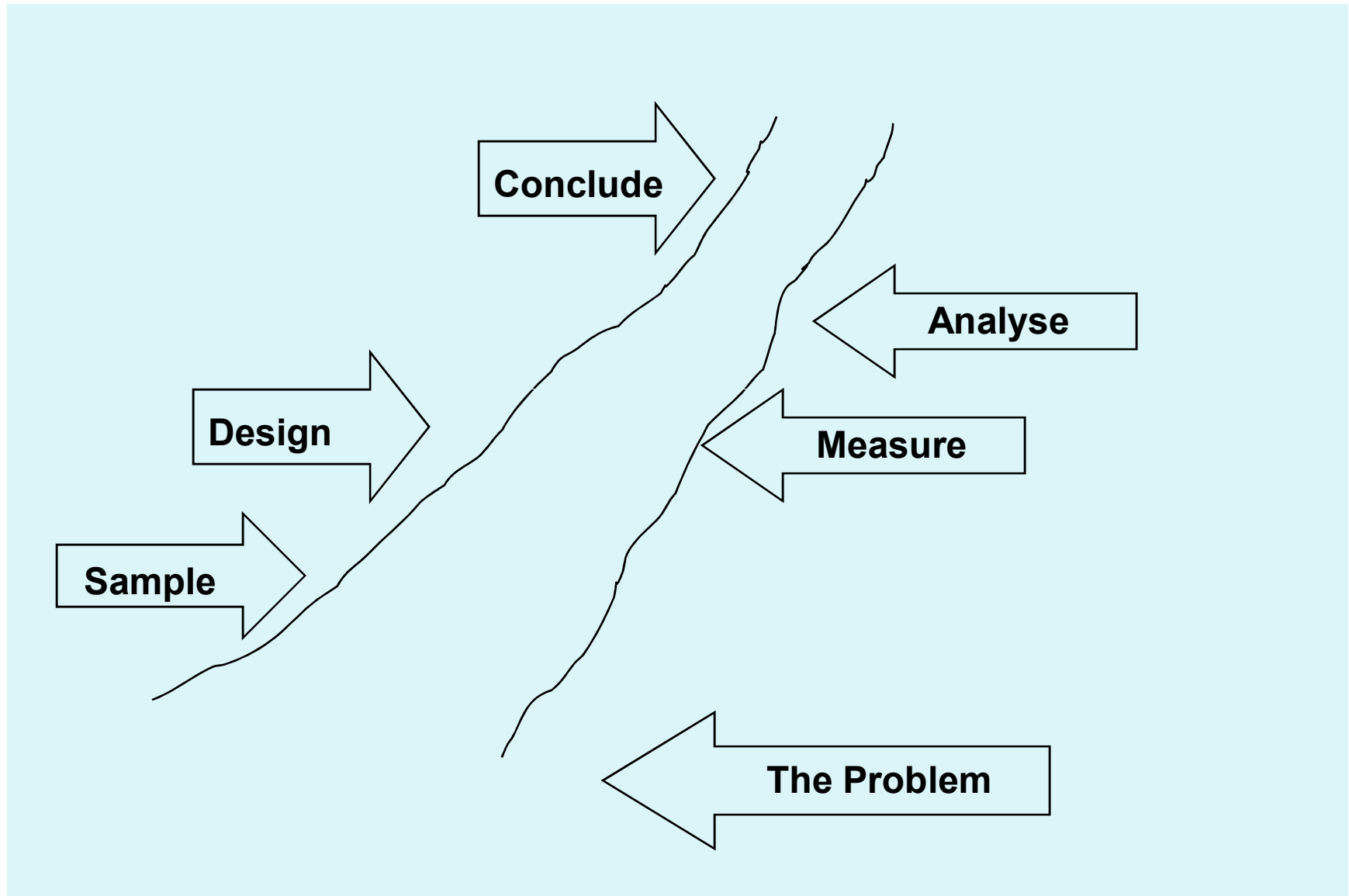
# Content

- What is a proposal?
- Preparation
- Structure of a proposal
- Why research proposal unsuccessful?

# In the early stage

- Identifying Research problem
- Formulating Research Question
- Formulating Research Objectives
- Literature review
- Experimental Design

# The Road to Research

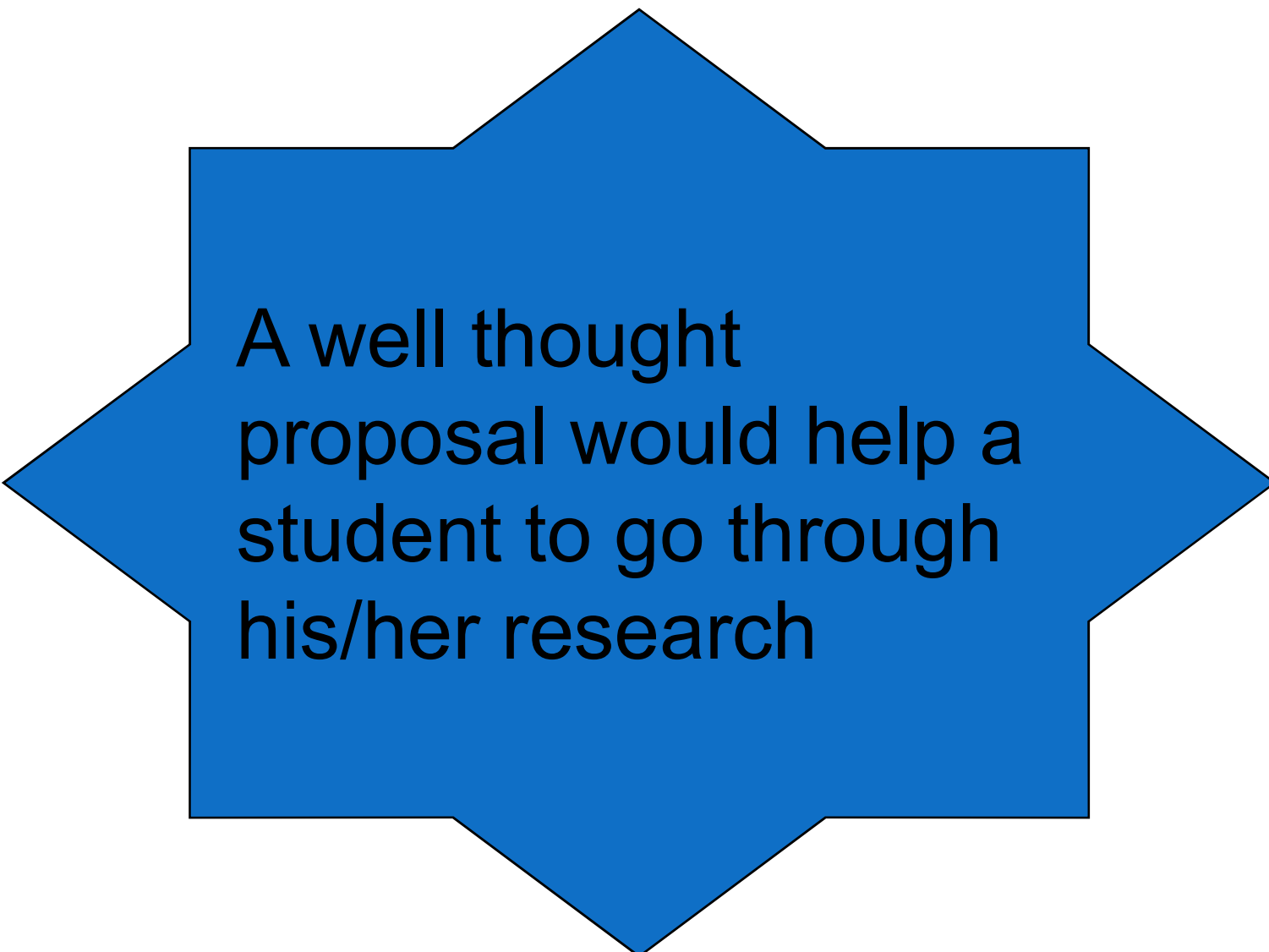


# Purpose of research proposal

- To make the reader to understand :-
  - What you are going to do
  - Rational of the research
  - Objectives of the research
  - Methodology
  - Expected output

# What is a proposal ?

- A good proposal should consists of the first three chapters of the thesis
- It should :-
  - begin with a statement of the problem/background information (Chapter 1)
  - A review of the literature (Chapter 2)
  - Defining of the research methodology (Chapter 3)



A well thought  
proposal would help a  
student to go through  
his/her research

# PhD Research

- More algorithmic
- Development of new technique
- Extension of existing new techniques
- Novel application

# Preparation

- Think about it
- Generate ideas
- Background reading
- Ask yourself
  - Am I familiar with other research that has been conducted in areas related to my research project?
  - Do I have a clear understanding of the steps that I will use in conducting my research?
  - Do I have the ability to go through each step?

# Structure

- Title
- Background to the problem or study
- Problem statement
- Objectives of research
- Scope and limitation of study
- Literature review
- Methodology
- Proposed schedule
- Significance of study
- References

# Title

- A good proposal has a good title
- It is the first thing that help the reader begin to understand the nature of work
  - Focused
  - Highlighting the main contribution of the research work
  - Use the keywords
  - Avoid ambiguous or confusing word

# Introduction

- Background study
- Problem statement
- Research questions
- Statement of research objectives
- Definition of terms

## Background study



A general review of the  
area of research

# Problem Statement

- Start with a general statement of the problem or issues
- Make sure the problem is restricted in scope
- Make sure the context of the problem is clear
- Cite the references from which the problem was stated previously.
- Provide justification for the research to be conducted
- Motivates to conduct the proposed research
- Highlight the problems/demerits of the available techniques

# Research questions

- Research questions would guide the proposed research into the perspective of the other research.
- The questions serve to establish the link between the proposed research with previous research.
- The research questions should show clearly the relationship of the proposed research with the field of study.

Most people associate research with doing something: conduct simulations, using equipment, or analyzing data.

---

However, the most critical parts of the research process are those parts that are associated with **thinking** not doing.

## Normally....

- Students want to do a project that is: '*...something about...*'
- You must turn that '*something about*' into a question.
- Posing a direct question will make the process of doing your research much more focussed. It will mean that your research consists of trying to answer the question.
- Distinguish between your 'research question' and what it is that you will research.

*A strong research idea should pass the  
“so what” test.*

Think about the potential impact of the research you are proposing.

What is the benefit of answering your research question? Who will it help (and how)?

- Be specific enough that operational definitions may be formulated later or the methodology, independent and dependent variables may be identified.
- Make sure the research questions provide a framework for reporting the results and discussion later
- Is it clear to the reader how the research questions arise from the issues and findings reported in the problem statement and later in the literature review?

- A good research question:-
  - Involves the search for relationships between two or more variables.
  - Is well defined and focused on specifics
  - Should be able to tell the reader what actually you are looking in particular.

# Literature Review

- Selecting Sources
  - ✓ Select literature that is relevant or closely related to the problem and purpose
  - ✓ Emphasize the primary sources
  - ✓ Use secondary sources selectively
  - ✓ Concentrate on scholarly research articles
  - ✓ Discuss your criteria for inclusion of articles

# Writing The Literature

- The literature should have an introduction, body and conclusion
- The introduction defines the framework of the review, the body that evaluates the literature and the conclusion summarizes the current state of knowledge on the problem

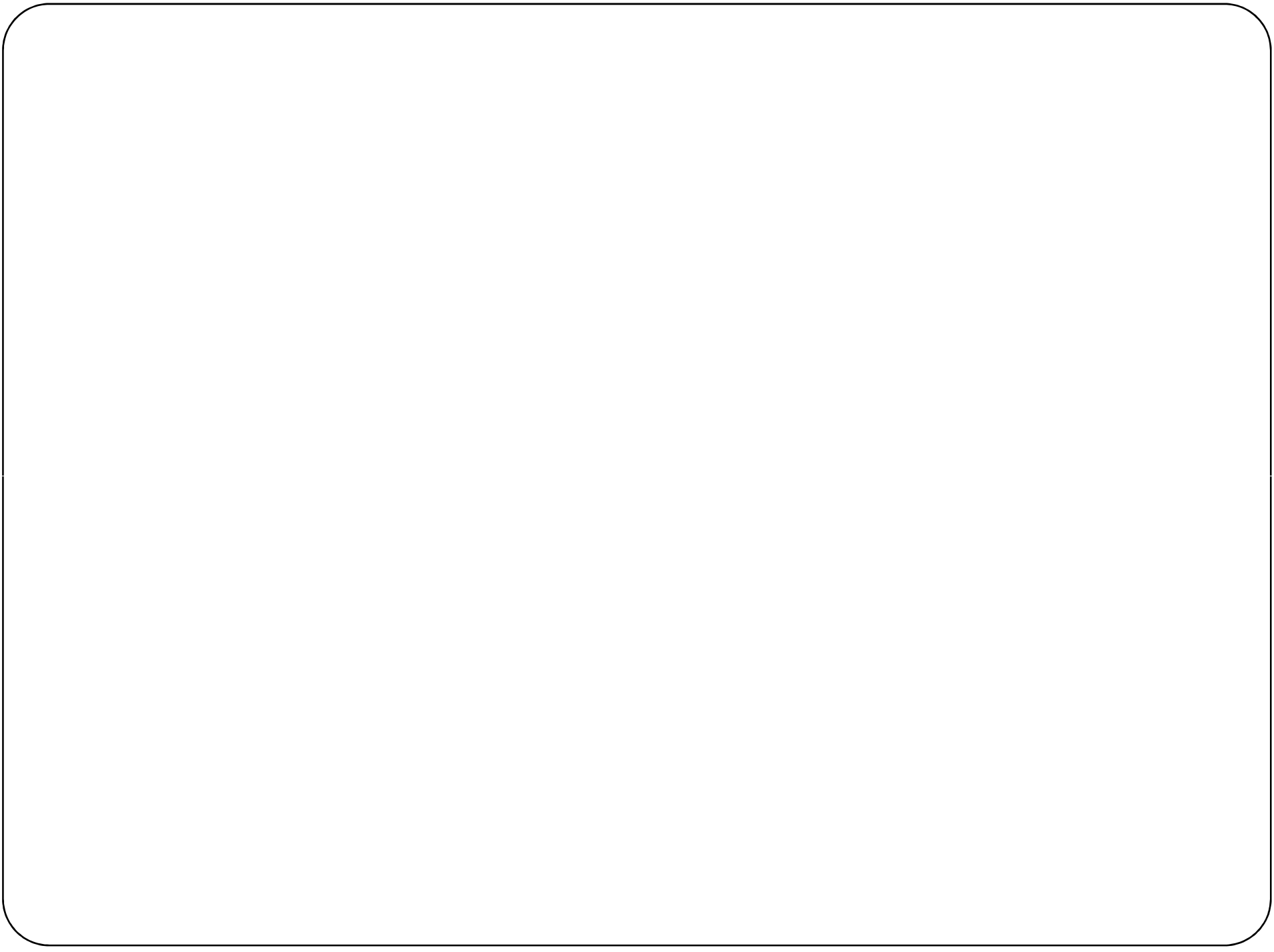
- Organize the review by topics or ideas, not by author
- Organize the review logically (least to most relevant – evolution of topic –by key variables)
- Discuss major studies/theories individually and minor studies with similar results or limitation as a group

- Adequately criticize the design and methodology of important studies so readers can draw their own conclusions
- Compare and contrast studies.
- Note for conflicting and inconclusive results
- Explicitly show the relevance of each to the problem statement

- Summary including a restatement of the relationships between the important variables under consideration and how these relationships are important to the hypothesis proposed in the introduction
- Identify the gaps in the current techniques that would be filled in by the proposed technique.
- Highlight the novelty of the proposed technique as compared to other existing techniques.

# Significant of research

- From the literature review, gap analysis can be conducted in order to see how the propose research would fill in the gap in the area of research.
- How does the proposed research relate to the existing knowledge in the area.
- Explicitly state the significance of your purpose or the rationale for your study. A significant research is one that:
  - ✓ Develops knowledge of an existing practise
  - ✓ Develops theory
  - ✓ Expands the current knowledge or theory base
  - ✓ Advances current research methodology
  - ✓ Related to a current technological issue
  - ✓ Exploratory research on an unexamined issue



# Scope and Limitation

- Provide the area/scope that will be considered in the proposed research and justification why it is being considered.

# Methodology

- Must related to the research objectives
- Highlight the breadth and depth of research
- Identify variables
- Research design – it would be good to put it into a flow chart
- Data collection plan
- Give a detailed sampling plan – the target population characteristics, specific sampling plan, target sample size

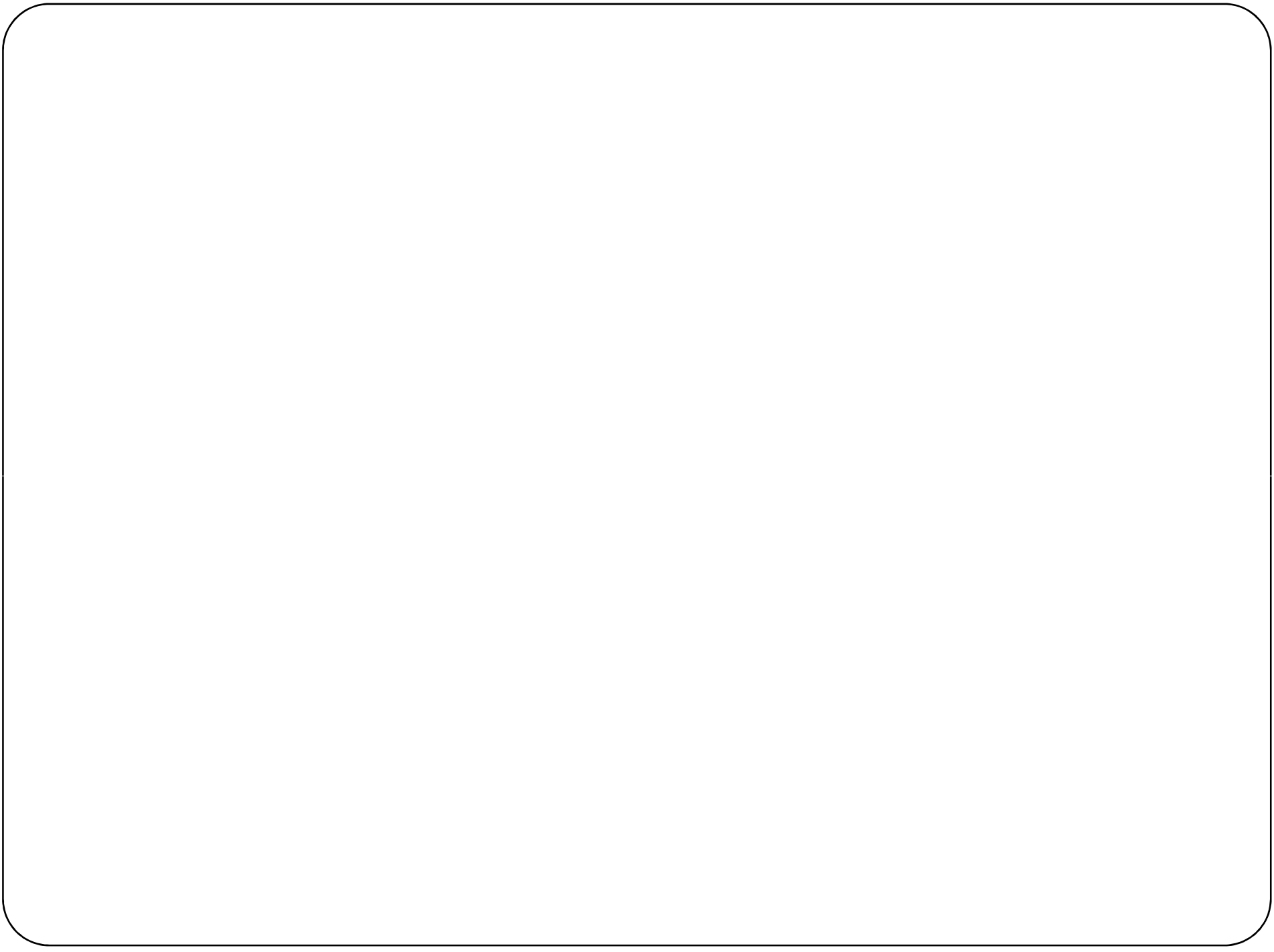
# Instrument

- Describe the instruments will be used to gather data (tests, techniques, surveys, etc)
- Provide reliability and validity information to show techniques are valid for the study
- Describe how the variables will be measured

- Provide justification for selection of instruments based on theory, research question, subject characteristics, etc.
- Provide published reliability of instrument and plan to establish reliability

# Procedure

- Describe how the study will be conducted
- When, how, where and by whom the data will be collected
- Describe the design of the test will be conducted or statistical test will be selected in this section.



# Anticipated Results

- Describe your anticipated results based on the literature review and theory based
- Write your conclusions if your research questions would be supported
- Write your tentative conclusions if your research questions would not be supported

# Reasons Why Research Proposal Are Unsuccessful

- The problem is of insufficient importance
- Purpose or demonstrated need is vague
- Problem is more complex than the propose realizes
- Research is based on hypothesis that is doubtful or unsound
- Proposed research based on conclusions that may be unwarranted

- Assumptions are questionable; evidence for procedures is questionable
- Approach is not rigorous enough, too naïve, too uncritical.
- Approach is not objective enough
- Validity is questionable, criterion for evaluation are weak or missing
- Approach is poorly thought out; methods poorly demonstrated

- Application is poorly prepared or poorly formulated
- Proposal is not explicit enough, lack of details, too vague or too general
- Rationale is poorly presented, logical processes not followed
- **Methods or procedures unsuited to stated objectives**

- The design is too ambitious or otherwise inappropriate
- Some administrative or practical problems are unsolved
- Unethical or hazardous procedure will be used
- The procedure is not well enough organized, coordinated or planned

- Some problems are not realized or dealt with adequately
- The overall design is unsound or some techniques are unrealistic
- The results will be confusing, difficult to interpret or meaningless
- Results from previous research are inadequate
- Proposer's knowledge or judgement of the scientific literature is poor

## Finally....

- The proposal can be used as first few chapters in the thesis
- Change the tense from future tense to past tense and then make any additions or changes so that the methodology section truly reflects what has been conducted

Thank you

Q & A